

BUILD THIS

LC METER

A low cost L/C meter that will give you lab-grade accuracy.

NEIL W. HECKT

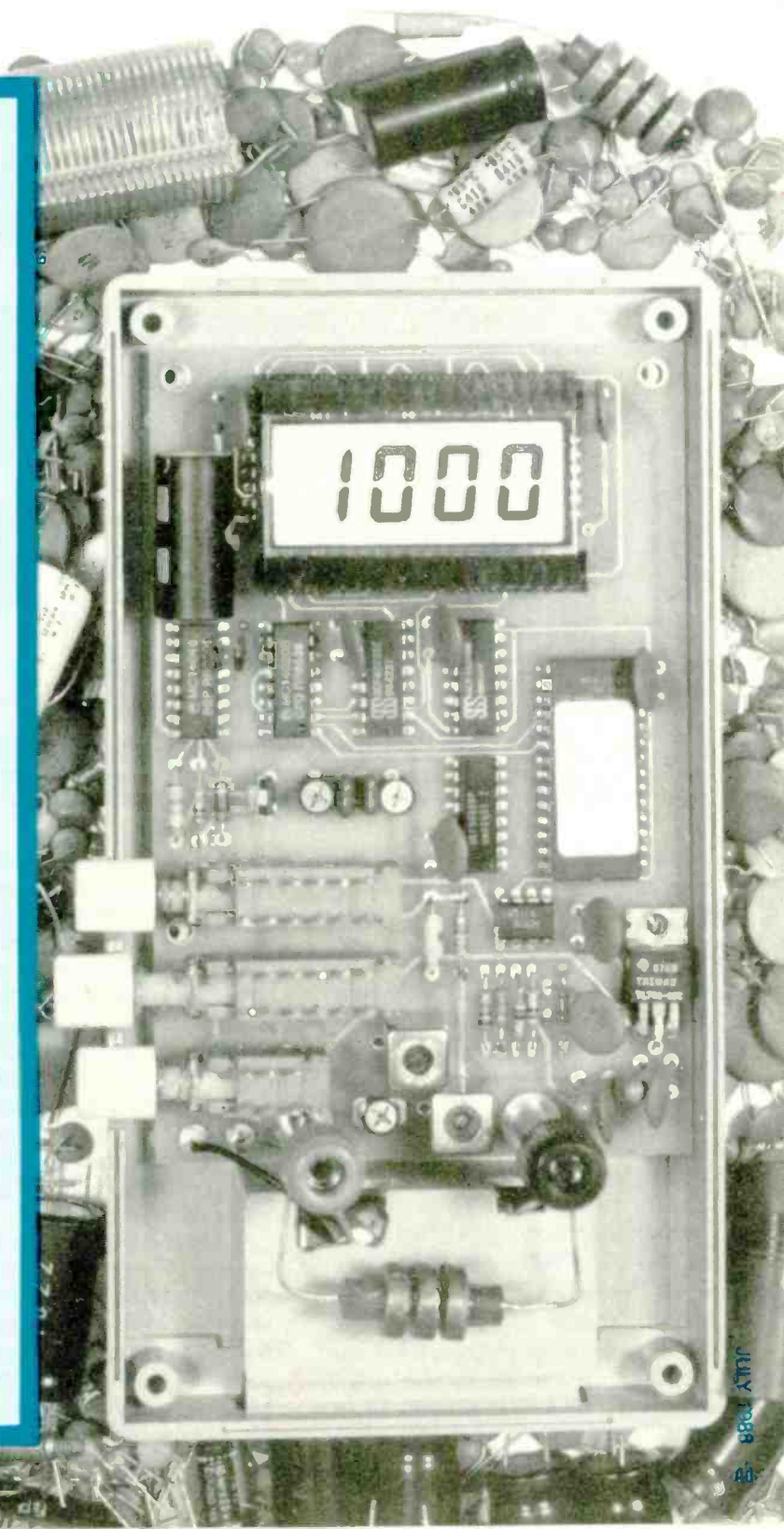
EXPERIMENTERS AND TECHNICIANS WHO WORK WITH RF filters, oscillators, and amplifiers have always needed an economical way to measure the value of small inductors. Generally, the measurement is made by inserting the unknown-inductor into an oscillator circuit having a known, fixed-value capacitor and inductor—a tank circuit. Because the capacitance and inductance are fixed values, a variation in the oscillator's output frequency is determined by the unknown-inductor. Since the oscillator's output frequency can be measured by a conventional low-cost frequency counter, the unknown-inductor's value can be easily computed by using an inexpensive hand-held scientific-type calculator. Similarly, unknown-capacitor values can be determined by again measuring how it affects the oscillator's output frequency.

Although the known-capacitor/inductor measurement technique gives reasonable accuracy for general and experimenter applications, unless extreme care is taken the end result is not what one usually accepts as "lab-grade precision." However, lab-grade accuracy can be attained by using a microprocessor-based measuring device having a built-in RF oscillator.

Whether the measuring device is a home-brew lash-up or microprocessor-based, both make measurements based on the frequency shift caused by the insertion of the component to be measured into the tank circuit of an RF oscillator. For RF circuits, that has the advantage of finding the *apparent* inductance or capacitance value as it would effect an RF tank circuit. The apparent value includes the parasitic capacitance of the inductor, or the parasitic inductance of the capacitor and its leads. That is unlike the typical capacitance meter, which inserts the component to be measured in a relaxation oscillator circuit that ignores inductance.

Calculating values

Where L_x represents an unknown inductance, and C_x represents an unknown capacitance, the formula for calculating the value of a component



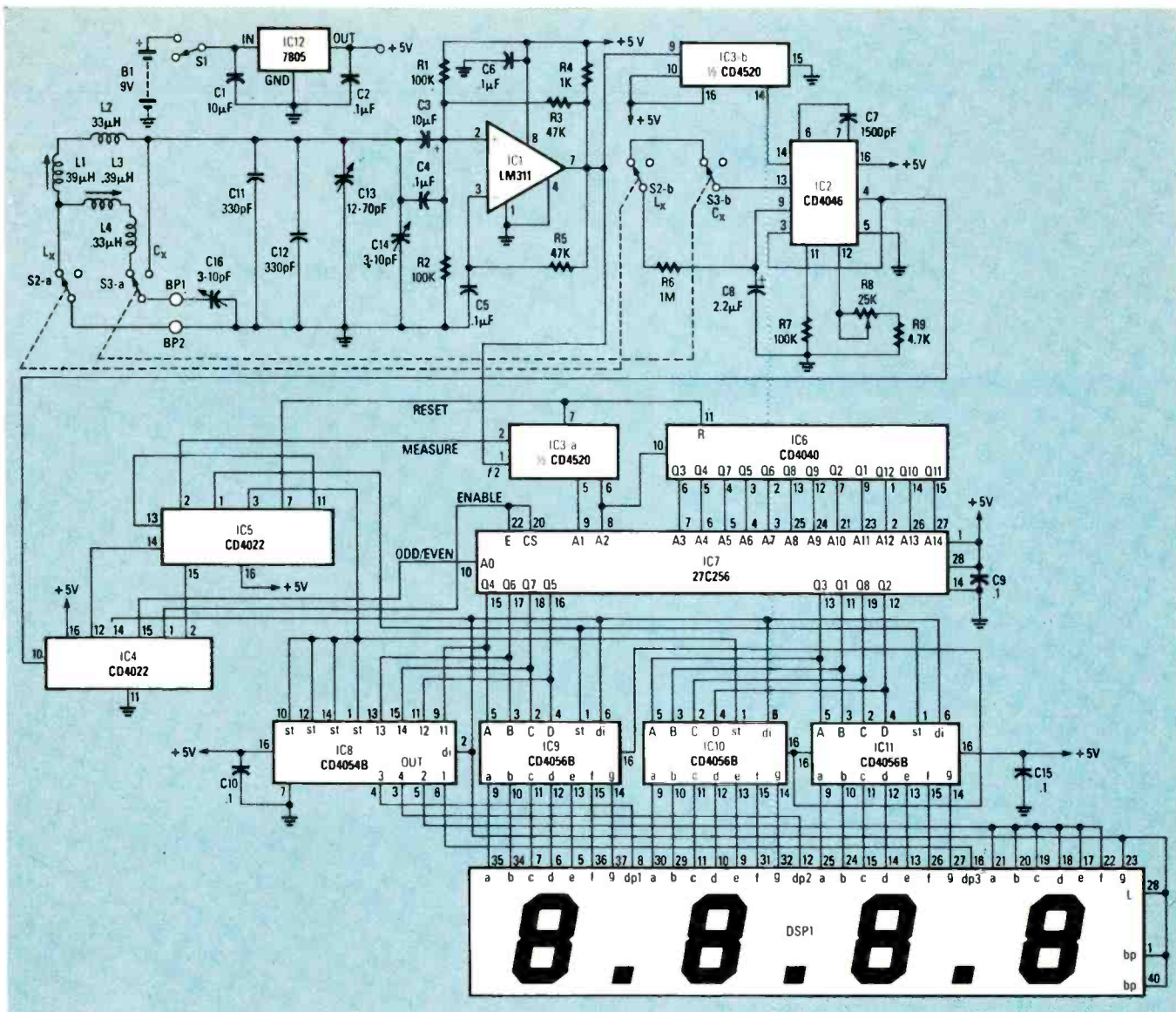


FIG. 1—THE RESULTS FOR ABOUT 15,000 EQUATIONS are stored in IC7, a pre-programmed EPROM. Input information is compared against the EPROM data.

as a function of the frequency shift it introduces into an oscillator tank circuit is:

$$L_x = ((f_1/f_2)^2 - 1) \times L_s$$

where L_s is the "standard inductor of the tank," or:

$$C_x = ((f_1/f_2)^2 - 1) \times C_s$$

where C_s is the "standard capacitor of the tank."

In both equations, f_1 is the oscillator's frequency without the unknown component; f_2 is the oscillator's frequency when the unknown component is installed in the oscillator's tank circuit.

Build a meter

Since there is a mathematical relationship between the known-component and unknown-component oscillator frequencies, to determine the

value of the unknown component we simply need to build some kind of device that measures the ratio between f_1 and f_2 , squares the ratio, subtracts 1, multiplies by the standard value, and then displays the result.

The first approach that comes to mind is the microprocessor device mentioned earlier. Unfortunately, a microprocessor-based instrument is relatively expensive because of its high IC count that includes a CPU, RAM, ROM, Timer, I/O, a display, and a power supply.

A lower-cost approach to a homebrewed L/C meter is a microcomputer that includes all of the functions of the microprocessor-based device in a single IC. Unfortunately, the initial investment would be on the order of \$40,000, which is a tad too expensive for a home-brew project. The newer

EPROM versions of the single-IC devices are reasonably priced, but require considerable expense in support equipment to develop, test, and program the software into the EPROM.

A more cost-effective approach for the hobbyist and technician is the device shown in Fig. 1; a simple frequency-counter circuit that drives an EPROM which is programmed with a large number of solutions to the L_x and C_x equations. In that way, all the mathematical functions, except for the frequency ratio, can be done outside the unit on a computer having the necessary capabilities. A 27C256 EPROM will hold 16384 solutions divided into two bytes per solution.

Circuit description

Since the frequency counter has a fixed resolution in cycles-per-second,

the EPROM contains solutions to the equations that are spaced apart by a predetermined number of cycles-per-second. Because resonant frequency is a function of the square root of inductance or capacitance, the size of the steps between adjacent solutions gradually increase, keeping approximately 0.1% (3 digit) resolution from 0.01 μ H to 10 mH, or 0.01 pF to 0.1 μ F. The advantage of that technique is that the unit will be auto-ranging.

Integrated circuits IC8 and IC9, the drivers for the liquid-crystal display (DSP1), are binary-coded decimal, requiring four bits per digit. Therefore, 12 bits are required for a three digit display. In order to provide the floating decimal point that's needed for the dynamic range of range of 10^6 , four bits are used to drive both the decimal

point and a fourth digit that is either zero or blank. Therefore, 16 bits are used to provide a direct readout in micro-henries. For capacitors, the displayed value must be mentally multiplied by 10.

The first problem was to design an oscillator that would always start, and which maintained oscillation even with very poor L/C ratios. Most oscillator circuits involve the impedance of the tank circuit in the loop gain, therefore, there are maximum values of L and C that can be used. Also, the normal output of an RF oscillator circuit is usually only a few millivolts, which requires amplification in order to drive the frequency-counting section of the instrument.

As shown in Fig. 2, the oscillator circuit uses a LM311 high-speed volt-

age comparator. Positive feedback from the output to the non-inverting input is provided by R3. The input is biased by R1 and R2 to approximately one-half of the 5-volt supply voltage. The tank circuit selected by switches S2 and S3 is capacitively coupled to IC1 by C3 and C4.

The best way to visualize the oscillator is to assume that IC1's output is a square wave. Driving the square wave into a parallel-tuned circuit through a large resistor (R3) allows the tuned circuit to filter out the sine-wave component of the square wave at the resonant frequency of the tank. (The sine wave will have a maximum amplitude when the square wave is at the resonant frequency.) Applying the sine wave to the non-inverting input of voltage-comparator IC1 produces a square-wave output of the same frequency. That, in turn, drives the tank circuit, resulting in an endless loop of oscillation. Thus, the oscillator runs at the frequency that produces the maximum sine-wave amplitude: the resonant frequency of the tuned circuit.

The inverting input of IC1 must be biased at exactly the same DC level as the non-inverting input in order for the oscillator to work. The R5/C5 RC filter from the comparator's output to the inverting input finds the average value of the square-wave output, which of course should be $\frac{1}{2}$ the peak-to-peak voltage, which is approximately $\frac{1}{2}$ the supply voltage. The same RC network is responsible for the starting and running of the oscillator. When power is first applied, the filter's output is zero, which causes IC1's output to be high (near 5 volts). The high charges C5 until its voltage reaches the bias level of the non-inverting input. That causes the output to swing low (near zero volts), putting a transient into the tank circuit, which causes it to ring at its resonant frequency. The ringing causes the output of the comparator to form a square wave of the same frequency which, after filtering by the tank circuit, is fed back to IC1's input to sustain oscillation.

The component to be measured, L_x or C_x , is always added to the standard inductor ($L1 + L2$) or standard the capacitor ($C11-C14$) in the oscillator's tank circuit. With the component under test installed, the oscillator's output frequency will always be lower than the frequency without the com-

PARTS LIST

All resistors $\frac{1}{4}$ -watt, 5%, unless otherwise noted.

R1, R2, R7—100,000 ohms
R3, R5—47,000 ohms
R4—1000 ohms
R6—1 Megohm
R8—25,000 ohms, trimmer potentiometer, 0.1" $\times$.2" spacing
R9—4700 ohms

Capacitors

C1, C3—10 μ F, 10 volts, tantalum
C2, C4, C5, C9, C10, C15—0.1 μ F, 50 volt, ceramic disc
C6—not used
C7—1500 pF, 100 volt, Mylar
C8—2.2 μ F, polystyrene (Panasonic ECQ-1225KZ)
C11, C12—330 pF, polystyrene or propylene
*C13—12–70 pF, trimmer capacitor (Mouser ME242-1270)
*C14, C16—3–10 pF, trimmer capacitor (Mouser ME242-2710)

Semiconductors

IC1—LM311N
IC2—CD4046
IC3—CD4520
IC4—CD4020
IC5—CD4022
IC6—CD4040
IC7—27C256 special programmed EPROM (see ordering note below)
IC8—CD4054
IC9—IC11—CD4056
IC12—LM7805CT
†DSP1—four digit LCD display, AND FE0202

Inductors

*L1—39 μ H, variable inductor (Toko 154ANS-T1016Z)
*L2—33 μ H (J. W. Miller 8230-56)
*L3—0.39 μ H (J. W. Miller 8230-10)
*L4—0.33 μ H, variable inductor (Toko BTKXNS-T1047Z)

Other components

B1—9-volt battery
*BP1, BP2—5-way binding post with 8-32 thread
*S1, S2, S3—DPDT alternate action switch (ITT Schadow 51281)
*3—pushbuttons for S1–S3
†2—LCD sockets
*2—8-32 $\times$ $\frac{3}{4}$ -inch threaded spacer with mounting hardware
*2—8-32 screws and star washers
*1—battery terminal clip !1—socket for DSP1, Samtec ESQ-120-12-T-S
1—enclosure, Pactec HPL-9VB

Note: The following parts and kits are available from Almost All Digital Electronics, 5211 117th St. SE, Bellevue, WA 98006.

A complete kit containing all components in the parts list with the exception of the EPROM, display kit, enclosure, and the PC board: \$69.95. A kit of hard-to-locate parts consisting of those indicated in the parts list with the * symbol: \$29.95. The programmed EPROM: \$19.95. The display kit consisting of those parts indicated in the parts list with a † symbol: \$14.95. The enclosure, with all holes machined and a front panel decal: \$19.95. The PC board with plated-through holes: \$19.95

A complete semi-kit (the switches are mounted and soldered) consisting of all of the above and a "standard" capacitor for calibration: \$149.95.

The completely assembled, tested, and calibrated unit: \$169.95. Add \$5 for shipping and handling per total order. Washington residents must add 8% sales tax.

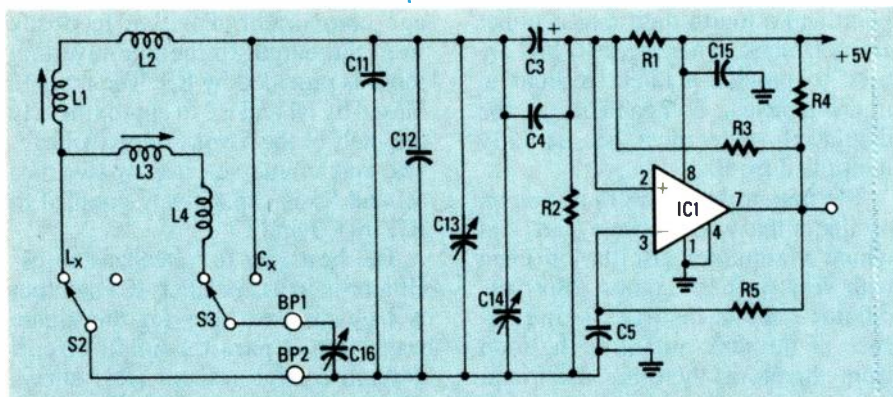


FIG. 2—A HIGH-SPEED VOLTAGE COMPARATOR, IC1, functions as an oscillator for both the capacitor and inductor tests. The unknown C and L values are determined by measuring their effect on the oscillator's reference frequency.

ponent installed. When the C_x switch (S3) is depressed, the component under test and zeroing-trimmer-capacitor C16 (which has been adjusted such that the sum of the stray capacitance + C16 = 10 pF) are connected in parallel with the C11–C14 tuning capacitors.

Similarly, when the L_x switch is depressed, the component under test and zeroing-trimmer-inductors L3 and L4 (which has been adjusted such that the sum of the stray inductance + L3 + L4 = 1 μ H), are connected in series with the L1/L2 tuning inductors.

As shown in Fig. 1, oscillator IC1 feeds a basic frequency counter consisting of counter-register IC3-a and IC6, which have a total of 16 bits; although only the last 14 are used ($2^{14} = 16384$). If a counter register is clocked by frequency f_1 , for a period of time T, the resultant value in the register will be $f_1 \times T$ cycles. If the measurement period is derived from f_2 as $T = N/f_2$ (where N is an integer multiplier, in this case a binary counter), then the accumulated count in the counter register will be $f_1/f_2 \times N$. In that way we have resolved measuring the ratio of two frequencies as required in the equation for L_x .

Compensating for drift

Frequency drift caused by temperature, or by aging of the components, is significantly suppressed if f_1 and f_2 are derived from the same basic oscillator. For example, suppose f_1 is 1 MHz and f_2 is 500 kHz, then $((f_1/f_2)^2 - 1) \times (L_1 + L_2) = (3.00 \times (L_1 + L_2))$. If the oscillator drifts 0.1% for both f_1 and f_2 , then $((f_1 + 1000)/(f_2 + 500))^2 - 1 \times (L_1 + L_2)$ also equals $(3.00 \times (L_1 + L_2))$. If, on the other hand, only f_2 drifted .1%

(which would be the case if f_1 had been derived from an independent oscillator), then $((f_1/f_2 + 500))^2 - 1 \times (L_1 + L_2) = (2.992 \times (L_1 + L_2))$, or 0.266% error.

There is a second, and even more significant advantage to this method of drift compensation. Regardless of the frequency, if f_2 is equal to f_1 , then $((f_1/f_2)^2 - 1) \times (L_1 + L_2) = 0$, thereby forcing an automatic zero function. Under the assumption that oscillator drift will be a percentage function rather than absolute, the effect of oscillator drift is canceled. Although that is not an entirely accurate assumption, it is reasonably valid because the majority of drift comes from drift in the values of L and C in the tank circuit. Those values drift as a percentage of their nominal value due to temperature effects, thereby leaving the accuracy of the instrument almost entirely dependent upon the values of L1 and L2 and C11–C14 (which can be easily adjusted to within 1%).

Although the foregoing explains the basic theory behind the counter section and its advantages, it is not quite the whole story. That basic concept could measure frequencies down to zero hertz (zero hertz would accumulate a count of zero in the counter register and f_1 would accumulate a count of 16383) just as a normal frequency counter would, and should do.

In order to make maximum use of the space in the EPROM, it is desirable to make the minimum value of the counter register correspond to the minimum value of f_2 (59.774 kHz) when $L_x = 10$ mH, or when $C_x = 0.1$ μ F, and the maximum value correspond to f_1 (713.3 kHz) when $L_x = 0$, or $C_x = 0$. By extending the range

of measurement period by "t" to $T + t$, the counter register will overflow back to zero after period "T" and continue counting for the period "t."

The final design has a period $T = 2^{16}/f_1$ and a period $t = 2^{12}/f_1$ such that the range of the counter register is 1024 to 17408 (a difference of 16384). To compensate for that, the bottom 1024 locations in the EPROM, 0–1023, contain the highest counts (16384–17408), and locations 1024–16383 contain counts 1024–16383. That is not the ideal correction for maximum EPROM usage, but it's easily achieved with simple circuitry. The 0.00 point is at location 17408 (1024) in the EPROM, while the minimum frequency of f_2 occurs at count 1455. There are, therefore, 431 wasted locations in the EPROM, or 15952 stored values in the range of 0.00–9990 μ H.

The frequency measured is $f_2/4$ (the counter register starts after the first two stages of IC3-a), so the formula to compute the accumulated value in the counter register for a given frequency of f_2 is:

$$\text{Count} = f_2/4 \times (2^{16} + 2^{12})/f_1 = f_2/4 \times 69632/f_1$$

$$\text{Count} = f_2 \times 17408/f_1$$

It is that count that addresses the EPROM to get the data for the displayed value. Maximum count corresponds to maximum frequency, which in turn corresponds to minimum L_x or C_x . The formula for the value of L_x for a given count is:

Solving the count equation for f_2 :

$$f_2 = (\text{count} \times f_1)/17408 = \text{count} \times 41.067$$

and since $L = 1/(4 \times \pi^2 \times f^2 \times C)$:

$$L_t (\mu\text{H}) = 1/(4 \times \pi^2 \times f_2^2 \times C_{11-C14})$$

and

$$L_x = L_t - (L_1 + L_2)$$

where

$$L_1 + L_2 = 70.5 \text{ mH and } C_{11-C14} = 705 \text{ pF}$$

$$L_x (\mu\text{H}) = 10^6/(4 \times \pi^2 \times 1686.52 \times \text{count}^2 \times C_{11-C14}) - (L_1 + L_2)$$

$$L_x (\mu\text{H}) = 213334.12 \times 10^6/\text{count}^2 - 70.5$$

$$C_x = L_x(\mu\text{H}) \times 10$$

Since the count value provides the EPROM address, and the L_x value provides the EPROM data, the last

formula and the pin-to-pin IC connections provides the necessary information to program the EPROM. The connections between counter register IC3-a/IC6 and the EPROM address pins may appear to be odd because they are optimized for a printed-circuit layout rather than the traditional pin assignments.

Similarly, the data bits of the EPROM are optimized for connection to the display drivers in the PC layout.

The translation for those connections is shown in Table 1 and Table 2. The "jumbled" connections are accommodated in the programming of the EPROM. A design problem which had to be overcome was that both frequencies f_1 and f_2 had to be available simultaneously to make the measurement. It was impossible to have the same oscillator run at two frequencies at the same time and the use of two different oscillators would negate the automatic-zero capability. Some means had to be provided to store frequency f_1 while the oscillator was operating at f_2 . The solution was a phase-locked loop, IC2.

Storing a frequency

The voltage-controlled oscillator of the phase-locked loop is adjusted by the frequency discriminator within IC2 until it is the exact frequency of the signal input. The voltage that tuned the oscillator to that frequency can be stored on filter capacitor C8, thereby essentially "storing" the frequency. The frequency can be stored accurately so long as C8's voltage does not change. The only thing that will cause the voltage to change is C8's shunt resistance. The input resis-

TABLE 2—EPROM DATA BITS							
4	400	200	800	100	1	8	2
8	7	6	5	4	3	2	1
C3	C1	B1	D1	A1	A3	D3	B3
128	64	32	16	8	4	2	1
C2	dp2	dp1	0 dig	dp3	A2	D2	B2
40	100	100	.10	10	10	80	20
stb A							
stb B							

tance of IC2 is specified as 10^{12} ohms, which shouldn't cause any shunt-resistance problem.

On the other side of C8 are the in-use open-circuit switch contacts of S2-b (L_X) and S3-b (C_X), which should also be many megohms. The most significant cause of discharge is C8 itself, which must, therefore, be an extremely low-leakage device. Ceramic, tantalum, and electrolytic capacitors are unsatisfactory; C8 should be a high-quality plastic capacitor.

The test circuit is such that depressing either the L_X or C_X switch will open the control loop of the phase-locked loop, thereby storing f_1 while simultaneously switching in the component under test into the f_2 oscillator for measurement.

The control section of the frequency counter is IC4 and IC5. The base frequency of the oscillator, f_1 , was first divided by 16 by IC3-b, and then divided by 512 by IC4 before clocking IC5. The period of the clock to IC5 is approximately 11.46 ms (87 Hz), however, IC5 is held in the reset mode by IC4 pin 2 for a period of 91.72 ms. After IC4 pin 2 goes low, IC5 advances from a count of zero to a count of one on the next positive transition of IC4 pin 12, which is 5.73 ms later. That makes the total measurement period $91.72 \text{ ms} (2^{16}/f_1) + 5.73 \text{ ms} (2^{12}/f_1)$, or 97.45 ms.

When pin 2 of IC5 goes low it inhibits the counter register (IC3-a/IC5), ending the measurement period. At that time EPROM IC7 is addressed with the result, which is a function of the component value L_X or C_X . Next, pin 1 of IC5 goes high for 11.46 ms, strobing the data from

IC7 into display drivers IC9 and IC11. At that time, pin 14 of IC4 is low, addressing the even addresses of IC7, which must contain the data for display digits one and three.

11.46 ms later, pin 1 of IC5 goes low and pin 3 of IC5 goes high, thereby strobing IC7's data into the digit-two driver IC10 and the decimal-point register (IC8). At this time pin 15 of IC4 is high, addressing the odd addresses of IC7, which must contain the digit two and the decimal-point data. An additional 11.46 ms later, pin 3 of IC5 goes low and pin 7 of IC5 goes high, resetting the counter register preparatory to acquiring a new measurement.

Another 11.46 ms later, pin 7 of IC5 goes low and pin 11 of IC5 goes high. The high is applied to IC5's ENABLE input, causing it to freeze at count four. IC5 remains in the idle mode until pin 2 of IC4 goes high 51.57 ms later (a total measurement/display period of 183.43 ms). The result is an update rate to the display of 5.44 times/second. An AC signal is taken from pin 14 of IC4 for display DSPI's backplane drive frequency.

Resident data

There are an infinite number of L and C values that will resonate at a given frequency. Determining the exact values which would simultaneously provide: a capacitance display exactly 10 times the inductance display, provide at least $0.01 \mu\text{H}$ per count in the frequency counter register up to $10 \mu\text{H}$, give a total count difference equal to or less than 16384 up to 10 mH with 0.1 mH per count,

continued on page 98

TABLE 1
BIT WEIGHTS
Counter Rom

1 →	2
2 →	4
4 →	2048
8 →	1024
16 →	8
32 →	16
64 →	64
128 →	128
256 →	32
512 →	256
1024 →	512
2048 →	8192
4096 →	16384
8192 →	4096

REACTS

continued from page 50

vate the water valves, and how many valves will be activated simultaneously. When you are calculating the power that is needed, remember to consider the amount of current that is used by the water valve's solenoid when it is first activated (the inrush current), which will be somewhat higher than the holding current of the solenoid.

In most cases, each of the octal I/O relays would activate only one valve. Therefore the relays, which are rated at 6 amps, will have no trouble at all, as the average consumer water valve requires less than 1 amp. Be sure to protect the transformer against shorts and overloads with the proper size fuse.

Also, follow the water-valve manufacturer's guidelines pertaining to the minimum diameter of the wire to be used, and their recommendations on waterproofing the wire and the connections. Finally, be sure to adhere to any local building codes that may apply to the installation. It's worth it in the long run.

For information on water valves, sprinkler heads, etc., contact your local lawn and garden supply store. Two major manufacturers of consumer sprinkler-system products are Toro and Rainbird. You should be able to find them, and many other manufacturers of such equipment, in the yellow pages.

In conclusion

By now, we hope that the wheels in your head have started turning as to what the REACTS modular system can do for you. It is no secret that computers are now being used for more than just number crunching, word processing, drafting, etc. The system we have been showing you makes computerized home control affordable. It also bridges the gap between process controllers and personal computers.

In the articles to come, we will discuss and build more of the process-control modules. We will also construct modules that will enhance the system, such as the battery-backed power supply and the CRT controller/printer interface. We hope you will stay with us.

R-E

LC METER

continued from page 45

and work with a simple binary counter scheme composed of inexpensive CMOS chips, required an Apple II program that ran for almost 4¼ hours.

The program that calculated all the required function addresses for the EPROM for each of the approximately 15,000 solutions to the equation ran for another 2½ hours. And after all the time spent calculating the EPROM data, the builder would need access to an EPROM programmer. To simplify everything, a programmed EPROM is available from the source listed in the Parts List.

Just in case you decide to test the EPROM data against your own calculations, bear in mind that the EPROM data have been compensated for the parasitic capacitive and inductance values that exists across the instrument's input terminals.

That's it for the theory. Next month we'll build and align the LC meter so that you can begin using it in your lab.

R-E

GPIB

continued from page 60

There are system and measuring-related constraints that also affect the data-transfer rate. Examples of those are high resolution (5 1/2 to 6 1/2 digit) voltmeters that use the very slow integrating A-to-D (analog-to-digital) techniques, precision low frequency counting, and narrow band spectrum analysis. Those slow real-time measuring processes were fine when a human could wait 5 or 10 seconds for a measurement to come up; however, with automated test instruments that make measurements within the millisecond range, that time frame is no longer acceptable. As a result, the test-equipment industry is converting to high-speed sampling techniques, burst measurements, smart peripherals, block-memory transfers, direct-memory accesses, and other such methods that allow the data throughput rate to be bound only by the transfer-rate of the computer-interface combination.

When we continue, we'll look at GPIB accessories and how to analyze and troubleshoot the bus.

R-E

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